The second computation rule for the flattened recursor: ap k of the path constructor is the given coherence path

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:flattening-rectnd-beta-ppt' (source lines 1989-1991, Lemma 6.12.8); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. Two context spans are retained: the author's auxiliary lemma 'thm:ap-sigma-rect-path-pair' with its proof, which the counted proof applies, and the author's own explanation of that lemma's shape.

Editorial note. The retained counter values (chapter 6, section 12, theorem offset 6) make the retained auxiliary lemma print as Lemma 6.12.7 and the counted lemma as Lemma 6.12.8, matching the original book numbering.

Editorial note. The six external references are navigation pointers to the results and displays the retained proof invokes by label, including equation (6.12.6) of the previous lemma; the counted proof is retained in full and nothing is replaced by them.

Editorial scope. One original result of the HoTT Book, counted once: the second computation rule for the flattened recursor, Lemma 6.12.8.

Allowed standard prerequisites (editorial).

- The non-dependent recursor of Lemma 6.12.5. From c : prd a:A C a -> Q and p : prd b:B prd y:C(f b) (c(f b,y) = c(g b, D b y)) one gets k : (Sigma w:W P(w)) -> Q with k(c-tilde'(a,x)) := c(a,x).. The lemma is stated in exactly that situation and its goal is the second computation rule for this k.
- Computation of ap on pair paths for componentwise maps. If k(x,y) := d(x)(y) then ap k (pairpath(s,r)) equals the displayed composite, provided full transport coherence is included; the statement is the author's auxiliary lemma.. It is invoked at source lines 1993-1994 with the four instantiations at lines 1995-1999 to reduce the goal.
- Computation rule of the induction principle for W. apd d (p(b)) is the composite of equation (6.12.6), passed across the equivalence of thm:dpath-arrow.. It is what makes the second factor at source lines 2005-2006 computable.

Source reference. Original lemma 6.12.5: sources/hott-hits/pinned-original-source.tex, source lines 1929–1929. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original lemma 6.12.1: sources/hott-hits/pinned-original-source.tex, source lines 1772–1772. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original equation 6.12.6: sources/hott-hits/pinned-original-source.tex, source lines 1941–1941. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original equation 2.9.4: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1324–1324. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original lemma 2.9.6: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1764–1764. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original lemma 2.3.5: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1902–1902. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Lemma 6.12.7. *Let $Y : X \to \mathcal{U}$ be a type family and let $k : (\sum_{(x:X)} Y(x)) \to Z$ be defined componentwise by $k(x,y) :\equiv d(x)(y)$ for a curried function $d : \prod_{(x:X)} Y(x) \to Z$. Then for any $s : x_1 =_X x_2$ and any $y_1 : Y(x_1)$ and $y_2 : Y(x_2)$ with a path $r : s_*(y_1) = y_2$, the path*

$$\mathsf{ap}_k(\mathsf{pair}^{=}(s,r)) : k(x_1,y_1) = k(x_2,y_2)$$

is equal to the composite

$$\begin{aligned}
k(x_1,y_1) &\equiv d(x_1)(y_1) \\
&= \mathsf{transport}^{x\mapsto Z}(s, d(x_1)(y_1)) && \text{(by (original lemma 2.3.5)}^{-1}\text{)} \\
&= \mathsf{transport}^{x\mapsto Z}(s, d(x_1)({s^{-1}}_*(s_*(y_1)))) \\
&= \big(\mathsf{transport}^{x\mapsto (Y(x)\to Z)}(s, d(x_1))\big)(s_*(y_1)) && \text{(by (2.9.4))} \\
&= d(x_2)(s_*(y_1)) && \text{(by } \mathsf{happly}(\mathsf{apd}_d(s))(s_*(y_1))\text{)} \\
&= d(x_2)(y_2) && \text{(by } \mathsf{ap}_{d(x_2)}(r)\text{)} \\
&\equiv k(x_2,y_2).
\end{aligned}$$

Proof. After path induction on s and r, both equalities reduce to reflexivities. □

At first it may seem surprising that Lemma 6.12.7 has such a complicated statement, while it can be proven so simply. The reason for the complication is to ensure that the statement is well-typed: $\mathsf{ap}_k(\mathsf{pair}^=(s,r))$ and the composite path it is claimed to be equal to must both have the same start and end points. Once we have managed this, the proof is easy by path induction.

Lemma 6.12.8. *In the situation of original lemma 6.12.5, we have* $\mathsf{ap}_k(\widetilde{\mathsf{p}}'(b,y)) = p(b,y)$.

Proof. Recall that $\widetilde{\mathsf{p}}'(b,y) :\equiv \mathsf{pair}^=(\mathsf{p}(b), q)$ where $q : \mathsf{p}(b)_*(y) = D(b)(y)$ comes from original lemma 6.12.1. Thus, since k is defined componentwise, we may compute $\mathsf{ap}_k(\widetilde{\mathsf{p}}'(b,y))$ by Lemma 6.12.7, with

$$\begin{aligned} x_1 &:\equiv \mathsf{c}(f(b)) & y_1 &:\equiv y \\ x_2 &:\equiv \mathsf{c}(g(b)) & y_2 &:\equiv D(b)(y) \\ s &:\equiv \mathsf{p}(b) & r &:\equiv q. \end{aligned}$$

The curried function $d : \prod_{(w:W)} P(w) \to Q$ was defined by induction on $w : W$; to apply Lemma 6.12.7 we need to understand $\mathsf{ap}_{d(x_2)}(r)$ and $\mathsf{happly}(\mathsf{apd}_d(s), s_*(y_1))$.

For the first, since $d(\mathsf{c}(a), x) \equiv c(a,x)$, we have

$$\mathsf{ap}_{d(x_2)}(r) \equiv \mathsf{ap}_{c(g(b),-)}(q).$$

For the second, the computation rule for the induction principle of W tells us that $\mathsf{apd}_d(\mathsf{p}(b))$ is equal to the composite (6.12.6), passed across the equivalence of original lemma 2.9.6. Thus, the computation rule given in original lemma 2.9.6 implies that $\mathsf{happly}(\mathsf{apd}_d(\mathsf{p}(b)), \mathsf{p}(b)_*(y))$ is equal to the composite

$$\begin{aligned} \big(\mathsf{p}(b)_*(c(f(b),-))\big)(\mathsf{p}(b)_*(y)) &= \mathsf{p}(b)_*\Big(c(f(b),\mathsf{p}(b)^{-1}{}_*(\mathsf{p}(b)_*(y)))\Big) && \text{(by (2.9.4))} \\ &= \mathsf{p}(b)_*(c(f(b),y)) \\ &= c(f(b),y) && \text{(by original lemma 2.3.5)} \\ &= c(g(b),D(b)(y)) && \text{(by } p(b,y)) \\ &= c(g(b),\mathsf{p}(b)_*(y)). && \text{(by } \mathsf{ap}_{c(g(b),-)}(q)^{-1}) \end{aligned}$$

Finally, substituting these values of $\mathsf{ap}_{d(x_2)}(r)$ and $\mathsf{happly}(\mathsf{apd}_d(s), s_*(y_1))$ into Lemma 6.12.7, we see that all the paths cancel out in pairs, leaving only $p(b,y)$. □